

MUNISING, MI, USA

WMO: 720198

Lat: 46.417N

Lon: 86.650W

Elev: 612

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54813

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.4	1.0	-12.2	2.9	-1.7	-8.1	3.6	1.8	28.6	17.7	26.9	19.6	9.2	180	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	85.4	69.9	81.9	68.2	78.8	67.0	72.6	80.8	70.7	78.8	68.9	76.1	10.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	112.5	77.3	67.8	104.6	75.1	66.0	98.1	73.4	36.6	80.2	34.9	78.6	33.4	76.3	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	22.2	19.4	-10.2	91.3	6.2	2.6	-14.7	93.2	-18.3	94.8	-21.7	96.2	-26.2	98.1		
(5)			-10.6	76.5	5.9	2.3	-14.9	78.2	-18.3	79.6	-21.6	80.9	-25.9	82.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	20.3	19.9	28.5	38.8	49.8	58.9	66.0	65.7	59.8	47.4	36.4	25.9
	DBStd	18.57	9.90	9.78	10.49	8.50	9.40	8.22	6.97	6.00	7.83	7.69	8.90	8.98
	HDD50	4239	920	842	671	353	123	13	0	0	8	145	417	747
	HDD65	8217	1385	1262	1131	785	482	219	75	65	196	546	859	1212
	CDD50	1777	0	0	5	18	117	280	496	485	303	65	8	0
	CDD65	280	0	0	0	0	11	36	106	85	40	2	0	0
	CDH74	1983	0	0	4	3	138	318	745	547	219	9	0	0
	CDH80	498	0	0	1	0	38	78	209	123	49	0	0	0
Wind	WSAvg	9.0	10.1	10.0	9.8	9.5	8.5	7.2	7.6	7.7	8.6	9.3	9.8	9.4
Precipitation	PrecAvg	35.8	3.0	2.0	2.1	2.2	2.9	3.1	3.2	3.3	3.9	4.1	3.1	3.3
	PrecMax	47.9	8.1	5.5	5.1	4.2	6.2	7.4	6.9	7.3	7.0	9.2	6.9	8.1
	PrecMin	25.7	1.4	0.4	0.5	0.3	1.1	0.5	0.6	0.2	0.6	1.1	0.7	0.7
	PrecStd	6.1	1.5	1.1	1.1	0.9	1.4	1.7	1.8	1.5	1.5	1.9	1.2	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	45.2	65.0	71.1	84.5	87.1	89.9	88.3	86.2	74.9	63.1	47.2
		MCWB	38.4	40.1	57.1	54.6	66.5	70.3	72.5	71.7	71.9	62.5	54.0	43.5
	2%	DB	38.8	39.8	52.5	64.6	77.7	81.8	85.6	83.2	79.9	68.3	56.2	42.0
		MCWB	35.8	35.3	44.7	50.9	63.0	67.1	70.3	69.2	66.9	60.0	49.7	39.5
	5%	DB	36.2	35.8	46.7	58.6	71.0	77.4	81.7	79.8	75.4	63.6	51.9	39.3
		MCWB	33.9	32.9	40.3	47.0	59.3	65.3	68.6	68.4	65.9	57.6	46.9	36.7
	10%	DB	33.9	33.3	42.0	52.7	65.5	72.8	78.0	76.6	72.1	59.5	48.6	37.1
		MCWB	31.9	30.6	37.0	44.4	55.3	63.0	67.4	67.1	64.4	54.7	44.2	34.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	41.4	57.4	57.4	70.0	73.2	76.2	74.4	73.6	65.7	55.5	44.8
		MCDB	40.8	44.9	65.3	67.2	80.9	84.0	86.0	83.0	82.9	68.2	61.9	46.8
	2%	WB	36.3	36.3	46.3	53.0	65.6	69.0	72.4	72.2	70.0	61.6	51.0	40.3
		MCDB	38.4	39.0	51.4	61.2	74.4	78.8	80.5	79.6	76.6	68.7	55.3	41.7
	5%	WB	34.2	33.0	41.0	48.1	61.3	66.7	70.6	70.2	67.6	58.3	47.4	37.2
		MCDB	36.0	35.3	46.0	57.4	69.3	74.7	78.7	77.4	73.4	63.6	51.2	38.8
	10%	WB	31.9	30.6	37.2	44.2	56.9	64.1	68.8	68.3	65.4	54.7	44.4	35.0
		MCDB	33.9	33.2	41.4	52.6	64.0	71.5	76.1	74.8	70.9	59.0	48.1	37.0
Mean Daily Temperature Range	5% DB	MDBR	11.5	13.8	14.6	15.2	17.9	18.0	17.6	17.3	17.0	13.9	12.0	11.0
		MCDBR	11.1	15.4	20.0	27.4	27.1	25.0	22.5	22.0	20.8	20.0	16.9	12.2
		MCWBR	9.9	12.1	13.4	16.5	15.6	13.9	11.7	11.4	10.9	13.1	11.5	10.7
	5% WB	MCDBR	10.7	14.3	17.7	24.4	25.1	21.3	19.5	17.8	17.4	18.2	15.3	11.0
		MCWBR	9.9	11.8	12.6	16.1	15.9	13.5	11.3	10.6	10.5	13.2	11.4	10.3
Clear-Sky Solar Irradiance	taub	0.268	0.281	0.312	0.347	0.375	0.385	0.390	0.376	0.361	0.333	0.303	0.269	
	taud	2.427	2.376	2.375	2.377	2.368	2.372	2.376	2.415	2.458	2.516	2.518	2.473	
	Ebn at Noon	268	285	291	289	283	279	276	276	271	263	252	255	
	Edh at Noon	22	28	32	36	37	37	37	34	30	24	20	19	
All-Sky Solar Radiation	RadAvg	340	607	1062	1412	1729	1872	1912	1619	1198	710	379	265	
	RadStd	27	82	99	146	124	143	98	103	82	83	47	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page